Asymmetric Temporal Reuse for Heterogeneous Components of Action-Chunked Robot Policies

***Abstract*— Repeated queries to an action-chunked policy produce several predictions for the same physical controller time, each conditioned on an observation from a different source time. Existing temporal ensembles and adaptive executors generally make one temporal decision for the complete action. We test whether heterogeneous action components can instead benefit from different source ages while holding the policy, current observation access, query cadence, and controller frequency fixed. Our training-free executor uses the current prediction for the six-dimensional arm command and the prediction generated 20 ticks earlier for the scalar gripper command. At 20 Hz, this is a 1.0-s source-observation gap, not a gripper hold. In a preregistered evaluation of one frozen ACT checkpoint on 126 paired LIBERO Object task-state blocks, this assignment achieved 80/126 successes (63.553/126 (42.1(49.2had positive paired-block and task-cluster 95% confidence-interval lower bounds, and every leave-one-task-out estimate was positive. A separate teacher-forced audit preferred old sources for arm translation, arm rotation, and gripper sign, showing that component prediction accuracy and closed-loop temporal-source utility are not equivalent objectives in this audited system.**

I. Introduction

Action-chunked robot policies predict several future controls from one observation. When the policy is queried again before a chunk expires, its predictions overlap: multiple queries propose an action for the same future controller time, but each proposal was conditioned on a different observation. This temporal redundancy supports smooth chunk execution, temporal ensembling, and adaptive selection among cached predictions [1], [2], [3]. It also exposes a basic execution choice: which observation-time source should control the robot now?

Most existing executors answer that question once for the complete action. Temporal ensembling uses shared weights across action dimensions, cached-action selection chooses one complete candidate, and adaptive execution chooses one prefix length or replan time [4], [5], [6], [7]. This shared decision is natural but not required by the policy interface. A robot action is heterogeneous. Continuous arm motion and an interaction command such as gripper opening can rely on different temporal information. A current arm prediction may benefit from the newest visual feedback while a gripper prediction from an earlier query may encode a temporally extended interaction decision. The latter is a possible interpretation, not an assumption of our method.

We isolate temporal-source age from action frequency. The frozen policy receives the current observation and is queried exactly once at every surviving controller step in every condition. Each query predicts a complete action chunk. At execution time only, we select the six arm dimensions and scalar gripper dimension from candidates that target the same physical time but were generated from different observations (Fig. 1). The gripper command can therefore change at every step even when its source query is 20 ticks old. This is not an open-loop gripper hold, a change in controller frequency, or a change in query cadence.

The study separates development from confirmation. A preregistered 2×2 developmental factorial did not support the generic hypothesis that mixed-source actions are less coherent: the balanced coherence estimate was $+.025$, with paired-block and task-cluster 95% intervals spanning zero. Its completed cells nevertheless showed a post-hoc directional pattern in which fresh-arm/old-gripper execution reached 62% success. That observation selected one fixed hypothesis. A subsequent preregistered study evaluated the selected fresh-arm/old20-gripper assignment on an untouched-state primary cohort and compared it with four frozen full-action baselines.

The confirmatory result is substantial in this audited system. The asymmetric executor achieved 80/126 successes (63.5%), versus 53/126 (42.1%) for newest, 55/126 (43.7%) for full old20, 62/126 (49.2%) for age-exponential ensembling, and 59/126 (46.8%) for tuned CogACT. The registered gains were 21.4, 19.8, 14.3, and 16.7 percentage points, respectively. Both paired-block and task-cluster interval lower bounds were positive for every comparison, as were all leave-one-task-out estimates.

A teacher-forced audit gives a deliberately different view. At the same source gap, demonstration prediction error preferred old sources for arm translation, arm rotation, and gripper sign. Closed-loop evidence instead preferred the fresh arm and old gripper. Thus a separable imitation loss can recommend a different source from closed-loop utility. Moreover, any arm-plus-gripper additive loss has an identically zero balanced 2×2 interaction, so it cannot measure the symmetric composition effect by construction.

This paper makes three contributions:

- We introduce a matched-query temporal-source factorial that assigns observation-time generations independently to heterogeneous action components while keeping the frozen policy, current observations, and one query per controller step fixed.
- We confirm a zero-training fresh-arm/old20-gripper executor that improves success over newest, full old20, age-exponential ensembling, and tuned CogACT on the preregistered primary cohort.
- We show that teacher-forced separable component error can disagree with closed-loop source utility and is struc-

turally unable to represent the balanced arm–gripper interaction.

The result is bounded to one frozen ACT checkpoint, LIBERO Object, one arm/gripper partition, and one fixed source gap. We do not claim a universal arm-fresh/gripper-old rule or optimality of the 20-tick choice.

II. Related Work

A. Overlapping predictions and temporal selection

ACT introduced temporal ensembling of overlapping chunk predictions for the same controller time [1]. CogACT's Adaptive Action Ensemble changes the shared weights on complete historical predictions using action similarity [2]. TAS instead caches actions produced from different temporal contexts and learns to select one complete candidate action [3]. Lazzati *et al.* show that delayed policies and implicit ensembles can explain why older observation-conditioned predictions help behavioral cloning in some settings [8]. These works establish that source time matters, but do not test whether heterogeneous components of one jointly predicted action prefer different source times.

B. Adaptive execution and stale-chunk correction

Several methods adapt a shared execution decision. AAC combines separately estimated translation, rotation, and gripper uncertainties to select one chunk prefix [4]. AutoHorizon uses model attention, PACE uses arm-motion phase boundaries, and HiPolicy uses multi-frequency predictions to change the execution schedule [5], [6], [9]. World-action-model executors compare prediction with realized progress to decide when the full chunk should be replanned [10], [7]. These methods change how long the robot acts before refreshing or choose one full-action candidate. Our controller and query frequencies remain fixed.

RTC freezes an unavoidable asynchronous prefix and inpaints the remainder of a new flow-policy chunk [11]. REMAC learns masked action correction, and A2C2 adds a small per-step residual from the newest observation [12], [13]. SEAM guides a new flow-policy chunk toward the previous chunk tail and can restrict that correction to selected action dimensions [14]. Dimension-selective correction is closely related technically, but it is not direct assignment of cached observation-time generations to action components.

C. Heterogeneous action structure and temporal routing

Action heterogeneity is itself established. ARP uses different chunk sizes for different action types in a learned autoregressive representation [15]. DAM-VLA specializes diffusion action models for arm motion and gripper manipulation, then coordinates them through routing and weighting [16]. Our novelty therefore does not rest on arm/gripper specialization. TRACT routes future action paths between current and next procedural phases and adds an arm-specific response-deficit correction [17]. Its temporal variable is future phase authority within a new chunk, whereas ours is the observation time that generated each component of the action executed now.

After a bounded primary-source search through August 24, 2026, we are not aware of prior controlled evaluations that independently assign temporal source generations to heterogeneous components of a jointly predicted action chunk. This is a bounded distinction, not a priority claim.

III. Method

A. Same-time candidates from different observations

Let $E_{u,q} \in \mathbb{R}^7$ denote the action for physical controller time u predicted by the frozen policy query at source time $q \leq u$. Repeated queries create a triangular cache of predictions. At physical time t, define the fresh candidate

$$F_t = E_{t,t} \tag{1}$$

and, for a fixed delay $d = 20$, the old candidate

$$O_t = E_{t,t-20}. \tag{2}$$

Both candidates target the same controller time; they differ only in the observation time that conditioned their source query. With a 20-Hz controller, $d = 20$ is a 1.0-s source-observation gap.

We partition each action as $a = [a^a, a^g]$, where $a^a \in \mathbb{R}^6$ contains relative translation and rotation commands and $a^g \in \mathbb{R}$ is the gripper command. The four factorial assignments are

$$\mathrm{FF}_t = [F_t^a, F_t^g], \qquad \mathrm{OO}_t = [O_t^a, O_t^g], \tag{3}$$

$$\mathrm{FO}_t = [F_t^a, O_t^g], \qquad \mathrm{OF}_t = [O_t^a, F_t^g]. \tag{4}$$

Before O_t is available, all conditions execute F_t. No candidate is averaged or smoothed in this factorial.

B. Fixed asymmetric temporal reuse

The confirmed executor is the fixed FO20 assignment

$$a_t^{\mathrm{FO20}} = [F_t[0{:}6], O_t[6]], \qquad t \geq 20. \tag{5}$$

It has no learned parameters and does not modify the policy. Every step still uses the current observation to issue one complete policy query and cache its $100{\times}7$ prediction. FO20 changes only which cached source supplies each component at execution. In particular, $O_t[6]$ advances from chunk offset 20 of query $t - 20$ to offset 20 of query $t - 19$ on the next tick. Therefore the executed gripper value may change every controller step.

The confirmatory baselines use the same query cache:

- **Newest**: execute F_t in full.
- **Full old20**: execute O_t in full once available.
- **Age exponential**: average every valid full-action prediction for t with shared weights $w_q \propto \exp[-0.03(t - q)]$ across all seven dimensions.
- **CogACT**: use the released cosine-similarity weighting with frozen $\alpha = 0.3$ and one shared weight vector for all seven dimensions [2].

The two weighted baselines test whether a strong shared full-action temporal rule explains the benefit. The study does not search over d, β, or α.

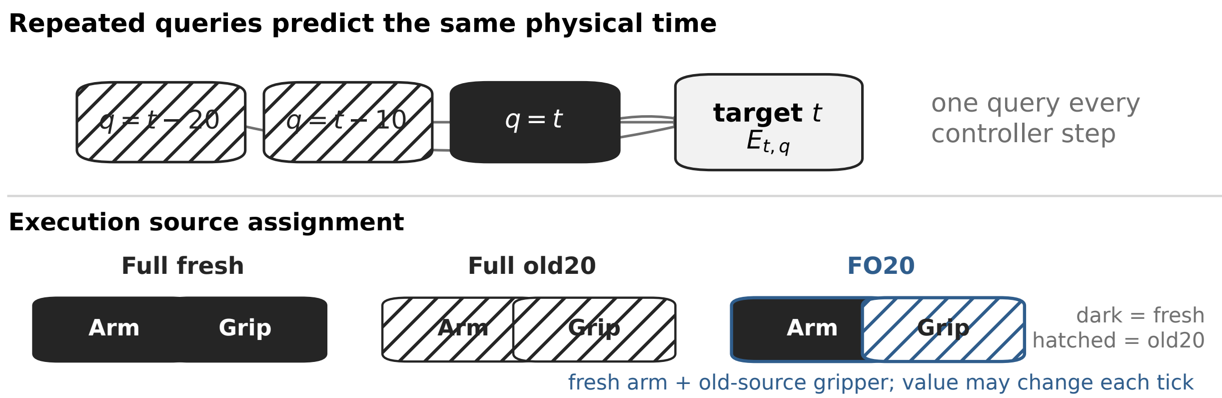

Fig. 1. **Temporal-source assignment at fixed query cadence.** Repeated policy queries predict the action for the same physical time t. The fresh candidate $F_t = E_{t,t}$ and old candidate $O_t = E_{t,t-20}$ are both available while the policy continues to observe and query every controller step. Full-fresh and full-old execution take both components from one source; the confirmed asymmetric executor takes the arm from F_t and the gripper from O_t. At 20 Hz, the source-observation gap is 1.0 s. It is not a 20-step gripper hold.

C. What a separable offline loss cannot measure

Consider any teacher-forced loss that separates by component,

$$L(a^a, a^g) = L_a(a^a) + L_g(a^g). \tag{6}$$

For a fixed target, write $A_F = L_a(F_t^a)$, $A_O = L_a(O_t^a)$, $G_F = L_g(F_t^g)$, and $G_O = L_g(O_t^g)$. The four losses are

$$L(FF) = A_F + G_F, \qquad L(OO) = A_O + G_O, \tag{7}$$
$$L(FO) = A_F + G_O, \qquad L(OF) = A_O + G_F. \tag{8}$$

Their balanced interaction is exactly

$$L(FF) + L(OO) - L(FO) - L(OF) = 0. \tag{9}$$

This algebraic identity does not imply that the four actions are behaviorally equivalent. It states only that an additive teacher-forced loss contains no arm–gripper interaction term with which to score symmetric composition.

IV. Experimental Design

A. Frozen system and execution contract

We evaluate a deterministic LeRobot ACT checkpoint [1] on the ten-task LIBERO Object suite [18]. The policy predicts 100×7 chunks for a 20-Hz relative-action controller with a maximum of 280 actions per episode. The policy's internal temporal ensemble and action smoothing are disabled. Table I summarizes the contract. All conditions receive the same current observations, preprocessing, complete policy queries, and postprocessing. Every completed episode satisfies one policy query per environment step.

TABLE I
FROZEN SYSTEM AND PRIMARY EVALUATION CONTRACT. A PAIRED BLOCK IS ONE TASK AND OFFICIAL INITIAL-STATE COMBINATION.

Item	Frozen value
Policy	LeRobot ACT; deterministic; no update
Checkpoint	340071d7…da9410
Prediction	100×7 action chunk
Partition	arm 0:6; gripper 6
Controller	relative action; 20 Hz; 280-step cap
Source gap	$d = 20$ ticks $= 1.0$ s
Query contract	one query per surviving step
Primary suite	LIBERO Object tasks 1–9
Primary units	14 states/task; 126 paired blocks
Outcome	binary task success

B. Developmental factorial

The developmental experiment used all four assignments in Sec. III on ten tasks and ten matched states per task, for 100 paired blocks and 400 episodes. Method order was randomized within each block, and the four methods shared the episode seed. The preregistered primary estimand was the balanced coherence contrast

$$C_{\text{coh}} = \frac{1}{N}\sum_i \left[\frac{S_{i,FF} + S_{i,OO}}{2} - \frac{S_{i,FO} + S_{i,OF}}{2}\right], \tag{10}$$

where $S_{i,c} \in \{0, 1\}$ is task success for paired block i and condition c. A positive value favors same-source execution after balancing the marginal arm and gripper assignments. Directional arm and gripper effects were examined only after the four registered cells were complete; they are developmental post-hoc evidence.

C. Teacher-forced component audit

The offline audit uses 82 demonstration episodes and 10,654 targets for which both fresh and old20 candidates exist. It evaluates normalized translation loss, normalized rotation loss, and gripper sign error against the demonstration action. The fixed cache, checkpoint, delay, and target set were defined without reference to the developmental closed-loop outcomes. The audit is descriptive because targets within an episode are repeated measurements and because Eq. (9) forces the symmetric interaction to zero.

D. Preregistered confirmation

The primary cohort contains LIBERO Object tasks 1–9 and 14 official initial-state IDs unused by prior project rollouts for every one of those tasks: $\{20, 21, 22, 23, 27, 31, 34, 35, 38, 39, 44, 45, 47, 48\}$. The identity audit did not inspect success outcomes. Each of the 126 task-state blocks received all five methods, yielding 630 primary episodes. Methods shared episode seed $330000 + 100\,\mathrm{task\_id} + \mathrm{state\_id}$ within each block, and a frozen random schedule with seed 20260831 permuted their order. A secondary 70-episode task-0 sensitivity was scheduled because task 0 had no untouched states; it is excluded from confirmatory inference. All 700 scheduled episodes completed with no exclusion or rerun.

The four registered comparisons are FO20 minus newest, full old20, age-exponential, and CogACT. The paired task-state block is the primary inference unit; controller frames are not independent success replicates. For each comparison we report the mean paired success difference, a 20,000-draw paired-block percentile bootstrap interval, a 20,000-draw task-cluster percentile bootstrap interval, and all nine leave-one-task-out estimates. Intervals use the 2.5th and 97.5th percentiles. A contrast was registered as stable positive only if both interval lower bounds and every leave-one-task-out estimate exceeded zero. These four planned comparisons form the complete confirmatory family. Intervals are contrast-wise rather than multiplicity-adjusted; the registered decision required the joint stable-positive criterion for all four comparisons. No outcome-driven method, cohort, or threshold change was made.

A preceding full-action control had established that temporal-source weighting changes closed-loop behavior in this frozen system. Its role is only to rule out an inert execution path; it is not a paper contribution.

V. Results

A. Generic mixed-source harm was not confirmed

The developmental factorial yielded 44/100 successes for FF, 40/100 for OO, 62/100 for FO, and 17/100 for OF (Fig. 2). Same-source cells therefore averaged 42.0% and mixed-source cells averaged 39.5%. The preregistered coherence contrast was only $+.025$. Its paired-block 95% interval was $[-.030, +.085]$, and its task-cluster interval was $[-.005, +.055]$. Both include zero. This experiment did not confirm a generic penalty for combining sources.

The directional pattern was analyzed after the registered cells were complete. The post-hoc fresh-arm marginal effect was $+.245$, with paired-block and task-cluster intervals of $[.155, .330]$ and $[.170, .330]$. The post-hoc old-gripper effect was $+.205$, with intervals of $[.140, .275]$ and $[.105, .305]$. FO exceeded OO on every task and was at least as successful as FF on every task, improving on eight and tying on two. These observations selected FO20 for confirmation; they are not an independent replication of it.

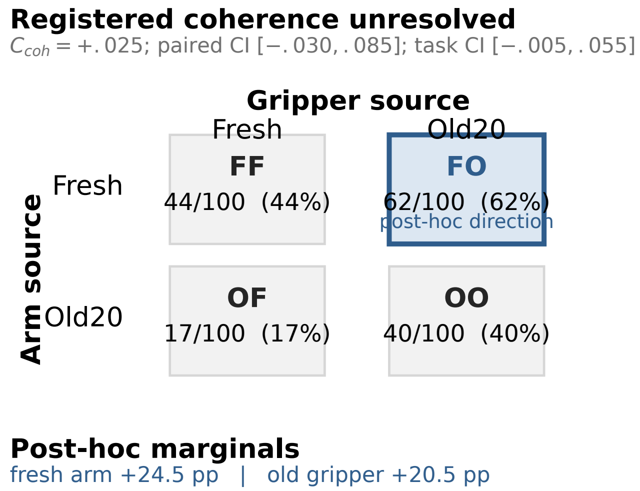

Fig. 2. **Developmental temporal-source factorial.** Cell labels give successes from $n = 100$ paired task-state blocks per condition. The registered coherence contrast compares the mean of FF and OO with the mean of FO and OF; its paired-block and task-cluster 95% bootstrap intervals both span zero. The highlighted FO direction and the marginal source effects were identified post hoc and used only to define the later confirmatory hypothesis.

B. Offline accuracy and closed-loop utility disagree

At $d = 20$, the teacher-forced audit preferred the old source for every measured component (Fig. 3). Normalized translation loss was .59578 for fresh and .50667 for old; normalized rotation loss was 1.12962 for fresh and 1.09877 for old. Gripper sign error was .30760 for fresh and .27400 for old. Lower is better for all three metrics. Closed-loop development instead favored a fresh arm and old gripper.

The disagreement is component-specific: the gripper direction matches across offline and closed-loop views, while the arm direction reverses. It does not show that demonstrations are wrong or identify a causal mechanism. It shows that teacher-forced component prediction accuracy and closed-loop temporal-source utility are not equivalent objectives in this audited setting. The observed offline balanced interaction is numerical zero to floating-point tolerance, as required by Eq. (9).

C. Asymmetric temporal reuse is confirmed

FO20 achieved 80/126 successes (63.5%) on the primary cohort (Fig. 4a). Newest achieved 53/126 (42.1%), full old20 55/126 (43.7%), age-exponential ensembling 62/126 (49.2%), and CogACT 59/126 (46.8%). Thus FO20 improved over both single-source controls and both shared full-action temporal-weighting baselines.

All four registered contrasts were stable positive (Fig. 4b). FO20 minus newest was $+.214$, with paired-block CI $[+.143, +.294]$ and task-cluster CI $[+.127, +.310]$. FO20 minus full old20 was $+.198$, with intervals $[+.095, +.302]$ and $[+.063, +.317]$. FO20 minus age-exponential was $+.143$, with intervals $[+.048, +.238]$ and $[+.016, +.278]$. FO20 minus CogACT was $+.167$, with

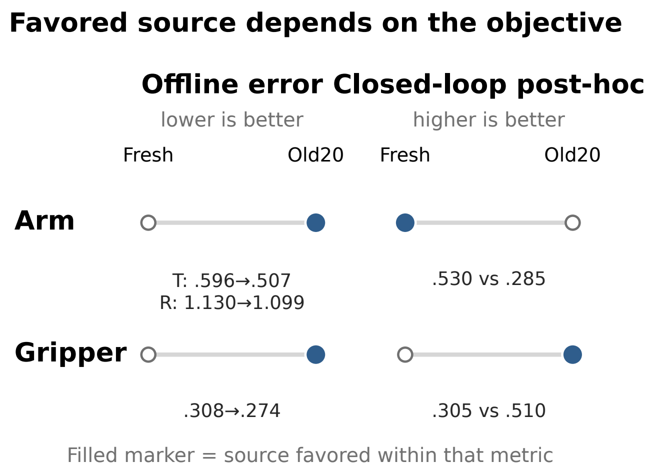

Fig. 3. **Temporal-source preference depends on the measurement objective.** Arrows indicate the favored source rather than placing incomparable raw quantities on one axis. Teacher-forced losses over 82 demonstration episodes and 10,654 eligible targets favor old arm and old gripper predictions. Post-hoc closed-loop marginal effects over 100 paired task-state blocks favor a fresh arm and old gripper. Lower error is better offline; higher success is better closed loop.

intervals $[+.071, +.262]$ and $[+.040, +.294]$. Every leave-one-task-out estimate was positive for every comparison.

The aggregate result does not imply uniform task-level improvement. Against full old20, for example, FO20 was lower on one task, tied on one, and higher on seven. The task-cluster intervals and leave-one-task-out checks explicitly carry this heterogeneity into the inference. The conclusion is therefore an aggregate, task-spanning effect for the frozen primary cohort.

VI. Discussion

The confirmatory study establishes a narrow but consequential execution result: components of one jointly predicted action chunk can benefit from different observation-time source ages. Because every method observes and queries at the same controller steps, the result cannot be attributed to different feedback frequency, query budget, or open-loop horizon. Because newest and full old20 use the same candidate cache, it also cannot be reduced to a general advantage for fresh or delayed actions. The benefit lies in the asymmetric assignment in this system.

This conclusion differs from the developmental question. The balanced factorial did not support a generic incoherence penalty for mixed-source actions. Its directional cell pattern motivated a new, fixed hypothesis, which the untouched-state experiment then tested. Keeping that distinction visible prevents the 62% developmental FO result from being counted as independent confirmation.

The result also clarifies how component-aware execution differs from related work. DAM-VLA already demonstrates that arm and gripper can merit specialized architectures [16]; our contribution is not heterogeneity alone. TRACT routes predicted future behavior across procedural phases and corrects arm execution [17]; our temporal variable is instead the past observation that generated a same-time cached prediction. TAS is the closest source-selection formulation, but selects a complete cached action [3]. These distinctions make the controlled intervention, not algorithmic complexity, the main methodological contribution.

The teacher-forced audit cautions against selecting a temporal source from offline imitation error alone. Old predictions better match demonstrations for both arm terms, yet the closed-loop factorial favors the fresh arm. Several explanations are compatible with this mismatch: fresh feedback may correct state deviation, demonstrations may reward temporal consistency differently from the closed loop, or the separable metric may omit behaviorally important coupling. The experiments do not distinguish these explanations. In particular, they do not prove retained intent, non-Markovian causation, or any general temporal-intent mechanism.

FO20 is deliberately simple. It adds no training, estimator, or policy head, and its result suggests a future adaptive executor could predict source utility per component. Such adaptation is not evaluated here. A learned rule would need an independent cohort and should be compared under the same current- observation and matched-query contract.

A. Limitations

The evidence uses one frozen ACT checkpoint and the LIBERO Object suite. It tests one six-dimensional arm/scalar-gripper partition and confirms one source gap, $d = 20$. The study neither proves that 20 ticks is optimal nor evaluates dynamic adaptation. It provides no causal proof that an old gripper prediction retains intent or that non-Markovian structure produces the success gain. There is no current confirmation on a second policy, another benchmark, a VLA, a dexterous or bimanual action space, or a real robot. The primary cohort spans nine tasks and paired unused states, which supports within-system credibility but not broad generalization. Finally, success is binary; logged kinematic and disagreement diagnostics follow treatment-dependent trajectories and are not used to make mechanistic claims.

VII. Conclusion

Repeated action-chunk queries provide multiple observation-conditioned predictions for the same physical time. In one frozen ACT/LIBERO Object system, assigning the fresh source to arm motion and the 1.0-s-old source to the gripper improved success over four full-action execution rules under identical observation access and query cadence. The result confirms that temporal-source age need not be shared across heterogeneous components of a jointly predicted action. At the same time, the offline audit shows that separable imitation accuracy can select a different source and cannot represent the symmetric component interaction. Component-specific temporal reuse is therefore a distinct execution variable that should be evaluated in closed loop, with its generality tested rather than assumed.

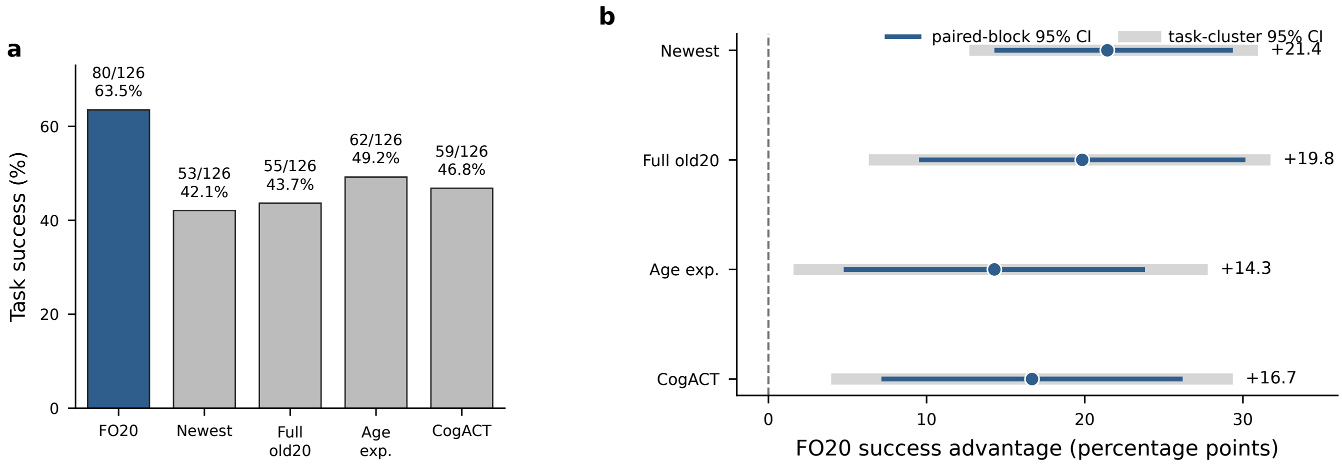

Fig. 4. **Preregistered confirmation on untouched primary states. a**, Success counts and rates for the five executors over $n = 126$ paired task-state blocks (nine tasks, 14 states each). **b**, Four registered FO20-minus-comparator success-rate differences. Circles are paired estimates; thick and thin intervals are 95% paired-block and task-cluster percentile bootstrap intervals, respectively, each from 20,000 draws. The dashed line marks no difference. All interval lower bounds and every leave-one-task-out estimate are positive.

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